

Maximise Performance Engineering with APM and Infrastructure Monitoring

New Relic Certification alignment

After completing this online workshop you will receive a **50% discount PROMO code via email** to use on any Associate- or Professional-level New Relic certification.
(Terms and conditions will be provided with the promo code and email)

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New Relic Certified APM Practitioner - Associate (APA)

[Online Workshop] Maximise performance engineering with APM and infrastructure monitoring

Monitor, analyze, troubleshoot, and improve the performance of modern applications & distributed architectures using APM and related data sets.

New Relic Certified APM Practitioner - Associate (APA) is for anyone with at least six months of experience in performance analysis using New Relic application performance monitoring.

This workshop contains content that aligns to the following exam sections highlighted in pink

Exam Guide

Section 1: New Relic Fundamentals Focuses on foundational knowledge, benefits and core features of APM	10%
Section 2: Managing Agent Data Focuses on instrumenting applications using APM agents, setting up application names, configuration settings, and tagging	20%
Section 3: Visualizing Insights Explores the tools and strategies for visualising and understanding APM data	30%
Section 4: Full Stack Observability Practices with APM Focuses on troubleshooting performance using New Relic APM. Addresses monitoring and management of applications and databases. Uses integrated data sets such as logs, traces, and infrastructure metrics; and shows how to set up alerts and manage notifications	40%



[APA Exam Prep Guide here](#)

[Certification Learning Path here](#)

[Register for Exam here](#)

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New Relic Certified Performance Engineer - Professional (PEP)

[Online Workshop] Maximise performance engineering with APM and infrastructure monitoring

Analyze and optimize systems and applications using a broad range of New Relic monitoring and performance management tools, with a focus on exceptional user experience.

The Performance Engineer - Professional (PEP) is for professionals who have 2 or more years of experience in performance analysis and optimization using New Relic monitoring and performance management tools.

This workshop contains content that aligns to the following exam sections highlighted in pink

Exam Guide

Section 1 New Relic Platform Capabilities and Data 30%
This section focuses on the fundamental capabilities of the New Relic Platform, including performance monitoring, error management, alerting, data visualization, and log management.

Section 2 Backend Application Performance 30%
This section delves into the monitoring and optimization of server-side applications, covering aspects like APM setup, microservices, and database performance.

Section 3 Client-side Performance 20%
This section concentrates on evaluating and improving the performance of frontend applications, using browser and mobile monitoring tools as well as synthetic testing.

Section 4 Infrastructure and Cloud Performance 20%
This section addresses the monitoring and management of server and cloud infrastructure, including cloud service integrations, container metrics, and database infrastructure.



[PEP Exam Prep Guide here](#)

[Certification Learning Path here](#)

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Agenda

- 01 New Relic APM overview
- 02 Lab: Explore the service summary page
- 03 Lab: Troubleshoot an APM performance issue
- 04 Lab: Troubleshoot an Infrastructure issue
- 05 Lab: Using Logs (time permitting)

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Lab: Explore the service summary page

The Service Summary page in APM is a set of curated charts to help you diagnose issues and follow links to delve deeper into related metrics, and troubleshoot the problem. Explore this page for a service and answer the questions below:

1. What is the significance of the four “score cards” at top of the page?
(Hint: Click the question-mark icons above each widget)
2. How can you change the time range displayed on the charts?
3. How can you view the data for individual instances of the application?
4. How can you see what other services or databases it is connected to?
5. How can you view the slowest transactions?
6. Extra credit: What is a transaction trace?

Lab: Troubleshoot an APM performance issue

Find the **Ad Service** in the **Demogorgon Prod** account. Set the time window to the past 60 minutes. Notice that there are several response-time spikes in the past hour

1. Based on the colors on the *Web transactions time* chart, where is most of the time being spent?
2. Which transaction(s) are responsible for the response-time spike?
3. Select the most time-consuming transaction and look at the transaction details. Do you notice anything on that page that may explain why the transaction is taking a long time?
4. Scroll further down the transaction detail page and select a transaction trace with a duration greater than 60,000 ms. Looking at the trace, what do you think caused the transaction to take so long?

Lab: Troubleshoot an Infrastructure issue

Scenario: The New Relic Pet Clinic application experienced a sudden increase in response time. Use Connected Infrastructure and APM to try to identify the problem.

Select the New Relic Pet Clinic application in the NRU Training account. Set the time picker to 24 hours, and locate a response-time spike (there should be one each day between 06:10 and 06:50 UTC).



Lab: Troubleshoot an Infrastructure issue

Use Connected Infrastructure and APM to answer the following questions:

- Were there any application deployments around the time of the incident?
- Was there an increase in application throughput that might explain the increased response time?
- Did response time increase across all hosts (containers), or only certain ones?
- Select an affected container and view the data for the associated host
(*Hint:* Related entities > See full map) Do you notice anything unusual about that host's data (*Hint:* Check host processes)

Lab: Explore Logs

1. Select the **NRU Demotron v4** account. From New Relic's main menu, select *Logs*
2. How can you see logs from the past 6 hours?
3. Filter the list to show only logs with a level of **ERROR**. How many logs are there?
4. Add a column to the results table to display the log level and `entity.name` of each log
5. How can you change your filter query to show only logs that contain a value for `entity.name`?
6. How could you save this view to quickly recall it later?
7. Click on a log entry and see its attributes. How can you easily share the details of a log entry with your team? What if they don't have access to your New Relic account?